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2016
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EMV+ 731 Ventilator by Zoll (formerly Impact Univent)

Quick facts:

-vent weighs aprox. 10 lbs
-air-worthied (for AE)
-can be used in CBRN environment (with gas mask canister)
-aprox. 10 hr battery life
-operates with or without supplemental O2: for FiO2 21-100%
-can be used on pt weighing ≥5 kg
-2 vent circuits: (make sure to also use appropriate sized HME filters and in-line suction)

Infant/ peds: 5-50 kg
Peds/ adult: ≥40 kg

-*Don't forget*: when you turn on the vent, it will not alarm for 1st 120 sec. (=monitor your pt!)

Equipment Test: (do NOT perform test attached to pt!)

1) Turn on (vent will do a self-test/ calibration)
2) After 120 sec. should alarm "pt. disconnect"
3) Press "MANUAL BREATH" and verify breath delivered
4) Occlude vent circuit, should alarm "High pressure limit"
5) Check battery status/ charging

Default parameters:

Mode: **A/C- Vol**, RR: **12**, Vol: **500**, PEEP:**5**, FiO2: **.21**, I:E **1:2.5** -can customize and save 2 additional profiles for start-up by:

MENU→ POWER UP SETTINGS →SAVE SETTINGS→ select USER 1 or 2 → CONFIRM

Add Pressure Support (to SIMV or CPAP):

-Press and hold "PIP" key, dial in PS

Low Flow 02

Ensure FiO2 is set to 21%

MENU→ select O2 RESEVOIR "ON"

Each shift:

-Check ETT cuff pressure with cuff nanometer (should be 20- 30cm H20)
-Check expired volume with Wright Spirometer (and compare it to the volume set)

Trouble shooting Ventilators Alarms "DOPE":

Displacement
Obstruction
Pneumothorax (tension)
Equipment

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Order of Circuit assembly:

1) **Circuit**
2) **ETCO2 detector**
3) **HME filter**

(Optional)

4) In-line suction (to ETT/ patient)

**Optional:* Add Bacterial/ Viral (B/V) filter to exhalation port if suspect aerosolized disease (TB, etc.)

**Optional:* Add debris/ B/V filter to intake port (in dusty environment)

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731 Series	AC 12, Vt 700, PEEP 5, I:E 1:2.5								
O2 Flow	0	1	2	3	4	5	6	7	8
FIO2	21	30	38	48	57	70	80	89	100
Vt(set)	740	732	725	718	711	703	691	689	682
Vt(actual)	700	692	685	678	671	663	651	649	642
% Chg	0	-1.1	-2.1	-3.1	-4.1	-5.3	-7.0	-7.3	-8.3
	AC 12, Vt 500, PEEP 5, I:E 1:2.5								
O2 Flow	0	1	2	3	4	5	6		
FIO2	21	30	43	56	69	89	100		
Vt(set)	527	523	514	506	502	493	486		
Vt(actual)	500	496	487	479	475	468	459		
% Chg	0	-0.8	-2.6	-4.2	-5.0	-6.8	-8.2		
	AC 18, Vt 300, PEEP 5, I:E 1:2.5								
O2 Flow	0	1	2	3	4	5	6		
FIO2	21	32	47	62	76	96	100		
Vt(set)	312	307	303	299	298	291	287		
Vt(actual)	300	295	291	287	286	279	275		
% Chg	0	-1.7	-3.0	-4.3	-4.7	-7.0	-8.3		

Note: Due to the slight difference between the densities of air and O_2 the tidal volume will decrease slightly as O_2 is entrained. The worst case is a <10% decrease in tidal volume when the entrained O_2 results in an FIO_2 of 100%. The tables below show both affect on tidal volume and the resultant FIO_2 for various O_2 supply rates:

Table 2: 731 Series - FIO2 at various O2 flow rates directly into the 3L Oxygen Reservoir